

K2-30b

R-2236

Benchmark run · workflow W8-benchmark-deep · record deep-bench_K2-30_230125 · created 2026-08-15T01:41:41+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459969.609 +/- 0.011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.129 +/- 0.044
Transit depth (Rp/Rs)^2	1.65 +/- 1.12 [%]
Orbital Inclination (inc)	85.0 +/- 2.35
Airmass coefficient 1 (a1)	0.7841 +/- 0.01
Airmass coefficient 2 (a2)	0.224 +/- 0.058
Scatter in the residuals of the lightcurve fit is	1.26 %
Best Comparison Star	#2 - [149, 256]
Optimal Aperture	3.6
Optimal Annulus	10.2
Transit Duration (day)	0.015 +/- 0.017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.129 ± 0.044 vs 0.1266 (deviation 0.04σ)
Duration fit vs published	0.015 d vs 0.0971 d (deviation 3.43σ)
Mid-time O–C	-26.7 min
Depth SNR	2.1
Residual scatter	1.26 %

Light curve

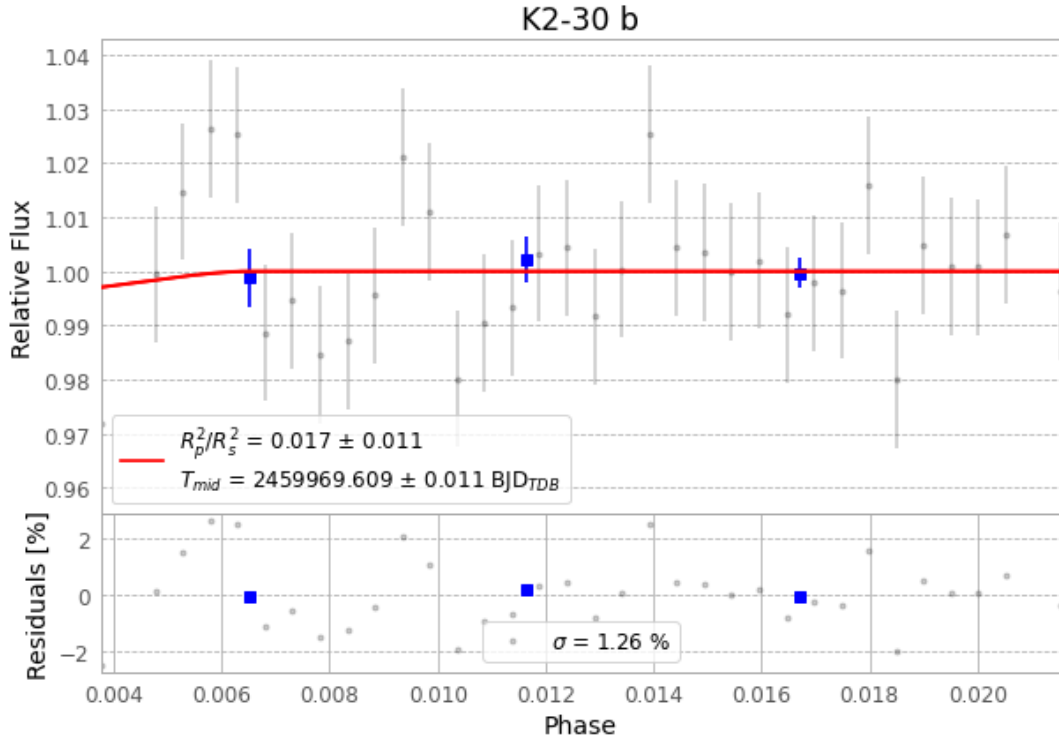


Figure 1 — FinalLightCurve_K2-30 b_2023-01-24.png (SHA-256 cd00f5967aef111c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [390, 235], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 03affd4a231f78e4...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

34 FITS frames (22 MB), K2-30230125025419.FITS → K2-30230125043916.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-K2-30-230125.log (6c844be0adbc...), FinalLightCurve_K2-30 b_2023-01-24.png (cd00f5967aef...), FinalLightCurve_K2-30 b_2023-01-24.csv (99ecd451eaeb...)

Compute resources

Wall time 107.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)